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Renewable Energy Landscape of Nepal Energy Systems in Nepal

The Advantages and Challenges for Adopting community-owned Renewable

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INTRODUCTION: UNDERSTANDING NEPAL'S CURRENT ENERGY SECTOR REALITIES

At present, Nepal is among the countries with the lowest energy consumption rates per capita in the world. Primary sources of energy utilized in Nepal can be categorized under three broad types (i) traditional (reliant on biomass fuels such as wood, agricultural residues and animal dung) (ii) commercial (reliant on fossil fuels and electrical grids), and (iii) alternative energy sources (e.g. new, renewable and non-conventional forms of energy such as micro-hydro, solar, wind power, biogas and briquettes). Biomass, hydropower and solar are the three major local energy sources relied upon by the population. Based on the fuel type, biomass provides 86% of the total energy consumed in the country, petroleum use (which is mainly consumed in urban areas) amounts to 9%, while grid electricity and renewables account for 2% and 1% of the mix, respectively (GoN, 2017). There is however, a shift underway as the national energy mix is expected to become more diverse in the future and increasingly dependent upon renewable energy.

In Nepal, 71.7% of households have electricity from the national grid, and 23% are connected to off-grid sources. Among households using an off-grid solution, the mini-grid and solar lighting systems of 10W, 20W and 50W respectively are the most common sources. 12% of Nepalese households are connected to an isolated mini-grid: a pico-hydro, micro-hydro, or mini-hydro system. Households that are connected to a pico-hydro system for their primary electricity source make up 2%. About 5% of the households in Nepal have no access to electricity in any form and rely on dry-cell batteries or solid fuels for lighting purposes (ESMAP, 2019).

To meet national energy demands, Nepal spends a large part of its GDP on electricity imports, especially from India in order to overcome the supply deficits (Poudyal et al. 2019). This energy imports from India has helped to meet the energy supply of the country and has aided to halt power cut problems that had crippled the country for many years. However, electricity that is imported from India is mostly coal-based. As a result, in the process of reducing the supply gap for electricity, Nepal is actively contributing to reliance on fossil fuel energy systems. This has, in effect, increased Nepal's global greenhouse gas emission footprint.

According to Nepal Electricity Authority's annual report of 2017, energy consumption in different economic sectors was as follows: residential areas consumed the highest portion of the energy i.e. 48% of the total energy, the industrial sector consumed 38% and the commercial sector consumed 12 %. The agriculture sector – where two-thirds of the Nepali population are engaged, consumed only 2% (NEA, 2017). The energy consumption in the transport sector was negligible, which indicated that this sector is under-developed, hindering the future economic development for communities that remain in remote areas.

A significant aspect of the recent energy history of Nepal relates to the crippling power cuts that Nepal faced for a decade from 2006-2016. These power cuts were credited to the supply and demand gaps due to the civil war and political instability at that time. However, upon further investigation, the root cause was actually structural corruption at the hands of the top bureaucrats of NEA in association with the industrial business interests mainly in the Terai region (in the border zones of Nepal). In this kind of corruption scam, industrialists bribed and used political influence and power to have constant access to the hydro-electricity, which cost around Rs.7/unit as opposed to relying on diesel power generators that would have cost them Rs.29/unit (Nepali Times, 2018). This saved millions of rupees for the industrialists, filled the pockets of corrupt bureaucrats but left the Nepali citizens in the dark for a decade.[1]

The price of electricity in Nepal for end-users is among the highest in the world (Poudyal et al. 2019). The power generation capacity has grown steadily in the country from 706 MW in 2011 to 856 MW in 2016 with an average of 3.9% yearly increment. However, the peak demand for electricity during the same period grew even faster, from 946 MW to 1,385 MW, or by an average of 7.9% yearly. This means there is a significant gap in the energy supply compared to the demand, which in fact is widening each year (ADB, 2017).

According tothe Government of Nepal's Economic Survey of 2019/20, 90% of Nepal's population have access to electricity. However, 10% of the total population still lacks access to modern and clean electricity. Most of this population is in the mountainous and hilly areas of the country, where the reach of national transmission and distribution networks remains limited. As a result, the remote areas where transmission line networks do not reach are mostly powered by micro hydro and solar-based mini-grids.

Similarly, Nepal's energy crisis has affected the people in rural areas more in comparison to the demography of those who reside in urban areas. The development of energy infrastructure such as the national grids and transmission lines in the urban area has been comparatively much better than rural areas. Approximately 80% of the rural population has not been reached by the national grid (Alex et al., 2016). These discrepancies in the development of energy infrastructures have posed a bigger problem in the rural areas of hills and mountains with regards to reliability and accessibility of energy.

The non-existence or lack of regular energy supplies has brought about major problems concerning the quality of health and education services in the rural population. People residing in rural areas are compelled to utilize low quality traditional fuels sources such as timber and animal dung for their daily energy purposes. This has adversely impacted the health of rural women who end up relying on burning traditional fuel in unventilated closed spaces in rural areas for cooking purposes. Around 7500 women die each year due to indoor air pollution created by the combustion of such traditional fuel (CRT/N, 2017). Likewise, children are forced to spend plenty of time in search of traditional fuel sources such as wood and timber as it is necessary for their families' livelihood. (Poudyal et al. 2019).

The intensity of the negative impact of Nepal's energy crisis on the marginalized communities is also disproportionately high. Nepal's marginalized communities, including indigenous peoples, Dalit communities, and religious minority groups among others, have faced a long history of systemic discrimination that has subjected them to unequal access to the state's resources and opportunities. This has created systemic inequality for such communities in all realms of society. This inequality also persists when it comes to the access of affordable and reliable energy. Besides, the impact of the current energy crisis has had more negative effects on such communities as they lie at the base of the economic pyramid due to poor income. These groups spend a large portion of their small income for energy purposes (Mary Robinson Foundation, 2015). Thus, when energy is inaccessible and unreliable, these groups are impacted the most due to the existing energy crisis in Nepal.

Access to electricity is highly dependent upon political influence when it comes to rural areas. The participation in decision making with regards to the energy system is often led by higher caste and dominated by male members. Since electricity is a privilege in the remote areas, the elite class residing in the remote areas often influence the governance and operation of energy systems. Meanwhile, the lack of political strength of leaders from remote hills and mountains is one of the major factors that have deterred the extension of the grid to these regions.

IMPACTS OF CLIMATE CHANGE ON THE ENERGY SECTOR

Nepal's energy generation and usage is dominated by hydropower, but climate change poses a serious threat to hydropower generation in Nepal. The climate change impact has not affected the energy sector but there is high likelihood the hydropower generation will be impacted by the change in climatic conditions as most of the hydropower in Nepal are based on run-of-river systems, so in turn, the climate change will exert pressure on snow cover, glaciers and change in river run off.

The government of Nepal has extensively pushed for the development of large-scale grid-based hydropower projects in Nepal to improve the supply of clean energy (IAP, 2018). Hydropower is categorized as a renewable energy source and at the same time, the country's geography presents numerous opportunities for its development. However, a thorough assessment of the impact of developing such mega-scale hydropower projects must be conducted and the recommendations of the EIA reports should be implemented. Construction of large hydropower projects will increase the supply of non-fossil fuel based energy but there are multiple costs that local community members and the environment must bear when constructing such projects. The government, donor agencies and the implementing partner need to consult with the community members of the project area and brief them about the project while addressing their concerns regarding the constructions of such large-scale projects.

The case of Tanahu Hydroelectric Project (THP), projected to generate 140 MW of electricity, in Tanahu district is one example of a hydropower project that has failed to adhere to the recommendations of the EIA done for this project. The hydropower project is funded by Asian Development Bank (ADB), Japan International Cooperation Agency (JICA) and European Investment Bank (EIB). ADB, JICA and EIB have signed a loan agreement of USD 150 million, USD 183 million and USD 70 million respectively to fund the project. With regards to environmental safeguards the hydropower project has conducted EIA (whose recommendations it has not completely addressed), drafted an environmental management plan and fish conservation management plan. Other social safeguards pertaining to development of this hydropower project includes a Resettlement and Indigenous Peoples Plan. However, according to ADB's official website, the THP has received an "A" for all three safeguard categories i.e. Environment, Involuntary Resettlement and Indigenous Peoples.[1]

The THP is a storage dam hence the reservoir will submerge large chunk of land, community forest, households, public infrastructures etc. Around 758 households will be affected by the project where 86 households will end up physically displaced and need relocation. Most of the households are owned by Majhis. Bote, Dunuwar and Darai people who are categorized as indigenous and vulnerable groups. Additionally, it is estimated that 62% of the community members will lose access to farmland that is essential for their sustenance, while 52% will lose access to community forests, grazing lands and the river, all of which are all essential for their livelihoods. About 51% will lose their source of income due to the impacts created by the project. An estimated 67% of the community members believe that they will not benefit from the project with the current implementation strategy. Though many of these affected households will get cash compensation, such indigenous and vulnerable communities lack experience on how to manage liquid cash and make investments in other lands. This ultimately leads them to spend all their money leaving them cashless and landless in the future.

When undertaking such development projects, meaningful participation of the indigenous and vulnerable communities in the consultation and the decision-making process is a non-negotiable prerequisite. A survey of the community members conducted by the International Accountability Project shows that 75% of the respondents were not consulted during the planning phase of the project, 93% were not given the opportunity to propose ideas for the development of their community and only 1% of the respondent said that their ideas and opinions were incorporated in the designing, planning and execution of the project (IAP, INWOLAG & CEMSOJ, 2017).

CURRENT ARCHITECTURE OF ENERGY POLICY AND PLANNING FRAMEWORK IN NEPAL

At present, two kinds of energy generation systems are predominantly adopted in Nepal (i) hydropower and (ii) utility scale solar photo-voltaic systems. Grid connected systems exist either as independent power plants (IPP) or are government-owned, where the government holds upwards of 60% of the total shares. All federal bodies involved in the process are under the Ministry of Energy, Water Resource and Irrigation (MoEWRI). Nepal Electricity Authority (NEA) is the sole off-taker, Department of Electricity Development (DoED) is responsible for facilitation and Electricity Regulatory Commission (ERC) is involved in policy formulation, planning and regulation of electricity generation, transmission and distribution. The head of all these institutions are politically appointed.

Besides these, there are local bodies like the District Coordination Committee and Municipality that facilitate and provide approval for use of local water sources, land and local committee formation[1]. The first step in developing a site for energy generation that will be connected to the grid is land acquisition, source identification and transmission line mapping. These activities are typically carried out by independent consultants. In case of hydro-river catchment area finalization, DoED seeks letter from Rural Municipality, which is based on obtaining a 'no objection letter' from District water resource committee. This process is required to determine if the community in the area is using the designated water source for themselves or not and if any other producers have claimed that part of the water body.

The second step involves power producers obtaining a survey license from DoED, valid for 2 years. This is followed by Detailed Feasibility Study (DFS), Environment Impact Assessment (EIA) and Grid Impact Study (GIA) which is required by DoED. Once the EIA is completed, NEA signs the Power Purchase Agreement (PPA) with the developer.

However, a major problem exists in the process of EIA evaluation. There is not a single time an EIA has been rejected so far by citing local demography, indigenous rights and/or resource depletion. Structural corruption exists in this process, which has led to the approval of all forms of EIA presented by the developer. After EIA approval, PPA and DFS are approved. DoED gives generation license to the developer and the whole process is monitored by DoED and Electricity Regulatory Commission.

The Budhi Gandaki project was initially intended to construct a hydroelectric dam to generate power in Nepal, but its development was halted in 2017. The reason for this was cited as 'instructions issued by parliamentary committees. There were also concerns about the potential impact on cultural heritage including the displacement of indigenous communities. It is important to note that the failure of proper EIA proposals or implementation can be linked to systematic corruption in Nepal.

Despite these concerns, the project's EIA was approved in 2018. It is worth noting that some court decisions have rejected environmental EIAs and called for further investigation into alternative measures to minimize the project's casualties.

Similarly, the Ministry of Environment, Science, and Technology (MoEST) rejected the proposal for a transmission line for the Upper Tamakoshi Hydroelectric Project in 2007 stating that it has ignored the basic principles of environment conservation, causing a delay in the project's initiation.

It is vital to prioritize the protection of the environment, wildlife, and local communities in the development of projects, especially those that could have a significant impact on the region. It is crucial to conduct comprehensive environmental impact assessments to address all concerns adequately. In this regard, it is essential to work closely with the local communities and incorporate their perspectives to minimize the negative impact of the project.

In the case of off-grid systems, under Ministry of Energy, Water Resources and Irrigation (MoEWRI) , the Alternative Energy Promotion Center (AEPC) is responsible for formulating policies, plans and programs, while rolling out subsidies for renewable energy technologies including solar mini-grids, micro-hydro etc. for off grid areas of Nepal. AEPC is mostly run by Government of Nepal funds and programs funded by donors. Current donors working with AEPC are as follows.

1. Asian Development Bank (ADB)'s "South Asia Subregional Economic Cooperation" program (SASEC)
2. ADB/United Nations Development Program (UNDP)'s "Renewable Energy for Rural Livelihood" (RERL) initiative
3. World Bank (WB)'s "Scaling up Renewable Energy Program" (SREP) for Biogas
4. Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)'s "Renewable Energy for Rural Areas" (RERA) project, and
5. UK Aid's "Nepal Renewable Energy Program" (NREP)

All the donor funded programs run in parallel with the AEPC. In the case of off-grid systems, policy, planning, implementation and focus areas are mostly controlled and administered by donor agencies. Besides AEPC and donor-funded programs, a few private companies install mini-grids in Nepal with their own equity investments, where tariffs are set by private companies themselves in coordination with user committees. However, they require permission from local government agencies to execute such projects. Local civil society organizations and community voices can't be seen anywhere in these process.

Concerningly, the government is primarily focused on prioritizing large-scale grid connected projects rather than off-grid energy planning processes, including those which would offer a transition from fossil fuel to clean renewable energy. As a result, in grid-connected areas, there will actually be an energy surplus by the end of 2021. Meanwhile, off-grid areas still lack access to energy systems where there could be actual engagement and ownership by - and of - the communities.

Surprisingly, Nepal is also exploring oil to meet its energy demand and for export purposes, which is against the Nepal's international commitments and a suicidal pathway of adopting carbon intensive energy generation. The Government of Nepal (GoN) in collaboration with China's Government has plans for oil drilling in Dailekh and other parts of Nepal, as identified by the Department of Mines and Geology[1]. There is a strong political push for oil drilling in Nepal, as the government is selling the story that oil drilling will address the present trade deficit, helping to make Nepal a prosperous nation from the revenue generated by exporting oil. However, Nepal needs to be committed to reduce fossil fuel dependency as this is already a goal mentioned in Nepal Climate Change Policy and second Nationally Determined Contribution commitments made by leaders. The Government of Nepal should therefore stop the plan of oil drilling and invest in community-owned renewables and pathways to economic development through a cleaner, just and sustainable future.

Under the "Multi-Actor Partnerships (MAPs) for Implementing NDCs with 100% Renewable Energy (RE)", the Alternative Energy Promotion Centre and its partner organization aims to introduce and consolidate multi actor partnership across varied stakeholders, government, civil society, business and academia with a vision of transitioning towards a 100% renewable energy scenario in Nepal. To support this, for example, Germany's Federal Ministry of Economic Cooperation and Development (BMZ) funded the preparation of provincial level Renewable Energy Plans for Gandaki and Bagmati.

LOOKING AHEAD: WHY SHIFT TO DECENTRALIZED COMMUNITY ENERGY SYSTEMS IN NEPAL

Nepal's Nationally Determined Contribution (NDC) has set a goal to expand clean energy generation from approximately 1,400 to 15,000 Megawatts, of which 5-10% will be generated from mini and micro-hydro power, solar, wind and bio-energy by 2030(GoN, 2020). Yet, the current system of energy provision adopted in Nepal has not been able to make energy provision reliable, affordable and accessible for all. With the widening energy supply and demand gap in Nepal and the ever-rising imports of fossil fuel from India to manage the crisis, the cost of energy consumption has remained expensive. Accessibility of energy in rural locations has been slow due to lack of national grid-connections. And the growing rates of fossil fuel imports has had devastating consequences on the national trade deficit and the environment while simultaneously aiding in the climate crisis (Poudyal et al. 2019). Hence, now would be an opportune time to switch to a more decentralized approach of energy production and consumption in Nepal.

Nepal's current energy system is heavily dependent on traditional energy sources and fossil fuel. This needs to change and must be replaced by clean renewable energy systems owned and operated by community groups. **There is a need to put people and communities at the heart of the shift towards clean energy. Decentralized approaches to energy in the form of community-owned energy projects will be a win-win situation in terms of social, environmental and economic benefits, as more renewable energy will be produced by people who use it, where they need it, and this will allow more citizens and communities to be involved in the energy transition process.**

Nepal should transform its energy supply system into a more sustainable system using clean and renewable energy resources, given the high costs of grid connection and the scattered population especially in remote areas (ADB, 2017). This transition is possible by giving ownership to rural communities for localized renewable energy projects as renewable sources naturally support decentralization since they are much more uniformly distributed and accessible in the remote areas where national grids are currently not put up due to their high cost of building.

In Nepal's context, there are many advantages to generating energy in a community setting. The renewable energy technologies in community-owned and managed energy systems will help foster social inclusion, as community members will be central to decision-making processes with regards to energy generation. It also has the potential to minimize the use of expensive and valuable resources for energy generation and transmission as energy will be produced remotely in the place where it is needed. Above all, it will allow for a reliable power supply for the community where energy will be distributed in an inclusive manner through local community groups.

Currently, Nepal's major policy guidelines such as the Article (51) G 3 of the Constitution and 14th five-year plan have promised the promotion of renewable energy generation in Nepal for reliable and affordable energy access. Similarly, Nepal has also pledged to fulfill the SDGs which aims to promote sustainable and modern energy for all while ensuring the accessibility of affordable, reliable, sustainable and modern energy for all. The promotion of community-owned energy systems will help to meet these national and international energy goals. However, with the current actors involved in the energy sector, these national policy guidelines and international pledges with regards to energy have been ignored -- and effectively put on the backburner.

Specifically, in the sector of renewable energy generation, large Nepali corporations, that are solely motivated by profits and endless economic growth rather than the provision of sustainable energy, have started to emerge as key players. Like many other sectors in Nepal, the entry and domination of profit hungry corporations have deterred us from achieving the intended essence of SDG -7 goal (i.e. to ensure access to affordable, reliable, sustainable and modern energy for all). Therefore, though corporations have invested money in the renewable energy sector, they have not adhered to making renewable energy affordable, reliable, sustainable and inclusive. For example, some of the large-scale renewable energy power plants controlled by major corporations, such as solar Photovoltaic (PV) plants are in the end utilizing productive arable lands of the Terai region. However, the use of such lands to generate renewable energy has reduced the agricultural productivity of the region. This practice has devastated the smallholder farmers of the region who rely mostly on agriculture for subsistence. Additionally, these corporate renewable energy plants have encouraged the practice of land grabbing that has destroyed numerous indigenous settlements and livelihoods of the local marginalized groups while their voices have been disregarded.

Economic incentives have attracted energy corporations to the renewable energy sector which is growing at an accelerated pace. However, their practices currently do not reflect their intention to promote the goals and objectives of the SDG and SE4All. Energy is not a commodity but a necessity in today's time, and its affordable access is everyone's right. The involvement of private entities such as solar companies in the energy market have turned energy into a commodity that they exploit in order to extract economic profits and influence the political landscape. The ownership of renewable energy projects by private corporations has prevented affordable accessibility of energy for all in Nepal.

Similarly, donor agencies with their current practices of implementing renewable energy projects have not been able to meet the sustainability criteria of national and international energy goals either. Currently, the projects they promote are designed for a fixed tenure with participation of limited stakeholders. This has narrowed the impact of donor-implemented renewable energy projects. Their current process fails to address the long term needs of the relevant stakeholders at the community level and their design does not address the issue of energy projects' sustainability post-project tenure. This has led to energy projects being initiated and operated under the supervision of the donors and implementing partners during the project tenure. However, once this tenure comes to an end, these energy projects are generally not sustained for continued energy generation because the roles of local authorities and community stakeholders have been insignificant from the very initial stages of project design. The 5 KW Solar Photo Voltaic mini grid installed in Marangbaru, Nayabasti Jahada rural municipality ward number 2 in Morang districts funded by UNDP is one such energy project that has not been sustainable after the completion of its purposed project tenure.

The short-term projects by donor agencies like Asian Development Bank's South Asian Sub Regional Economic Cooperation (SASEC), UNDP's Renewable Energy for Rural Livelihoods (RERL) and World Bank's Mini-grid Energy Access project have lacked long term vision in terms of project sustainability, community-centered approaches and inclusivity. An example of such energy project is the 5 KW Solar PV mini grid in Morang district funded by UNDP mentioned in the previous paragraph. Instead, donor institutions just focus on the technical aspects of the energy project. If donor agencies want to help improve the energy sector of Nepal, they must adhere to democratic values for long-lasting and impactful results. The donor agencies must involve all community stakeholders from the beginning regarding the financing, design, operation, maintenance and longevity of energy projects developed. An alternative viable approach would be when collaboration among the donor agencies, private sector and community members can take place, enabling the operation of a successful community-owned energy project -- where communities ultimately have control over the energy systems they use.

POSSIBILITY OF CIVIL SOCIETY INVOLVEMENT IN DECENTRALIZED ENERGY IN NEPAL

The Alternative Energy Promotion Center (AEPC) established in 1996, has been the national focal point for the promotion and development of renewable energy. It functions independently and is under the Ministry of Energy, Water Resources and Irrigation, Government of Nepal. AEPC developed “National Renewable Energy Framework” (NREF) in 2017, which is an umbrella mechanism that combines and coordinates policies and programs in the renewable energy sector behind a set of strategic objectives. AEPC is the cornerstone institution for policies and programs related to renewable energy in Nepal, and is tasked to bring together the federal government, development partners, and other relevant stakeholders including CSOs, private sector and financial institutions in the energy sector to align with the provision of the NREF under its leadership (NREF, 2017). It is important to note, however, that the number of civil society organizations that are primarily working in the energy sector of Nepal are limited in number, and are mostly comprised of large, well-established organizations and INGOs. These organizations are the major CSOs that are consulted by policy makers while formulating and implementing energy related policies, plans and projects in Nepal. However, such organizations have often prioritized their own interests while developing renewable policies.

The emphasis by such organizations on community-owned, operated and managed renewable energy systems is minimal – despite their proven effectiveness to tackle the problem of inaccessibility of energy in countries like Nepal. On the other hand, the number of homegrown, local CSOs predominantly engaging in the energy advocacy landscape of Nepal is scarce. As CSOs whose primary engagement differs from the national energy sector, they are marginally involved in shaping the energy discourse of Nepal. In fact, such local CSOs’ meaningful participation has been limited in stakeholder consultations when it comes to designing renewable energy polices, plans and projects of Nepal -- despite their continuous engagement in the energy landscape at the grassroots levels.

Having said this, the role of local CSOs in formulating NREF was non-existent. There was no multi-stakeholder consultation conducted while designing, planning and executing any large scale energy plans and projects with the local CSOs. The AEPC was mandated to encapsulate all relevant stakeholder’s concerns while formulating renewable energy projects. However, CSOs that could have raised key concerns from the perspective of community members in rural and remote areas were systematically left behind in the process. In order to effectively implement changes to promote community-owned renewable energy projects, CSOs that advocate for the decentralized community-owned energy in the renewable energy sector will need to be included as major stakeholders in formulating the country’s renewable energy policies. The experience of local CSOs working in the energy sectors at grassroots level have the potential to guide the renewable energy policies of Nepal to a more community-based approach, since they are the ones who are most aware of the grassroots problems and concerns relating to energy reliability and accessibility in community settings.

For example, Digo Bikas Institute (DBI) and its partner organization in Dhapsung village, Helambu Rural Municipality, Sindhupalchowk District have built a 16KW community-owned solar micro-grid managed and operated by Dhapsung Women’s Group. The motivation behind such initiatives is to promote democratic, participatory, open, accountable, decentralized community-owned energy projects in Nepal.

DBI also aims to provide documented evidence on the viability of such community-owned energy projects as a possible alternative approach to aid the current energy problems and display the way in which solar related interventions are being carried out in Nepal to meet the SDGs 7 and 13. The execution and field experience while implementing such community-owned renewable energy system could be of great value in the execution and policy level to decide the course of the decentralized renewable energy generation. If there could be increasing involvement of local CSOs in the process of drafting energy policies from the initial phase, the energy policies would capture a broader perspective on justice and inclusiveness for the relevant communities. This will help to voice the demands of community members and further allow renewable energy projects to be more community centric which would lead us in the direction where energy could be reliable, affordable, accessible and the energy projects would be sustainable for a longer period of time.

COMMUNITY SOLAR MICRO-GRID IN NEPAL

Solar micro grid projects are a highly feasible renewable energy source for the remote hilly areas in Nepal where grid access is not possible due to rugged geography and terrain, and the extension of transmission and distribution networks is near impossible. The installation of the community solar micro/mini-grid is generally administered by Alternative Energy Promotion Center (AEPC). However, various other actors like CSOs, private sectors and local governments play direct and indirect roles in development of solar micro grids in Nepal.

The majority of solar micro-grid projects are largely funded by subsidy models in Nepal where AEPC and donor agencies provide 90% subsidies for the system and the rest is collected from community contribution. Another prevailing modality for renewable energy generation is the Public-Private Partnership (PPP) model, which is gaining popularity among donor agencies. An example of such PPP model that is promoted by the GoN in collaboration with World Bank (WB) to encourage private sector and commercial financing is the private sector led Mini Grid Energy Access Project (MGEAP). Through MGEAP private entities and cooperatives will be mobilized to provide electricity in rural areas as “Energy Services Companies” or ESCOs. 40% of the project cost is funded by AEPC whereas the private entities invest the remaining 60%. The role of ESCOs will be to provide technical expertise and financing capacity to develop, build, own and operate renewable mini grids projects in Nepal. Furthermore, various non-profit organizations provide system and cash grants to install solar micro grids in Nepal. Besides this, private companies invest, own and operate micro grids in remote villages for monetary benefits.

Among all the above-mentioned types of systems in practice in Nepal, the least beneficial type with respect to the tariff structure are the privately owned micro grids where people are charged up to Rs.40/unit (highest among the tariffs paid anywhere in Nepal) followed by the PPP model type energy projects conducted under AEPC and donor-funded nonprofits. AEPC and donor-funded projects generally have sustainability issues as the projects are often phased out after its completion due to the lack of meaningful participation of community actors in the projects. Similarly, complicated governance issues arise in such projects where community ownership and handover of the system is not transferred smoothly at the end of project tenure. Such problems leave the communities dependent on local governments and private companies in cases of technical problems. Additionally, a major crisis occurs when the system’s batteries need to be replaced after the completion of its warranty period. This is because the tariff collected from the energy project alone is not adequate to provide for the battery replacement cost, as it is usually more than 55% of the total cost of the system.

Additionally, the energy projects under AEPC and donor-funded agencies currently are stringent with regards to handing over ownership of the energy projects from the initiation stages to local communities. As a result, they end up withholding decision-making powers from community members. Although there is some level of community engagement currently in the form of local committees, these do not facilitate the needed engagement of communities in independent decision-making. It is not enough to simply implement the practice of decentralized energy generation with democratic values in words only. In order to promote community energy systems the community members must be given maximum autonomy with regards to the decision-making process of the energy project. Thus, energy projects supported by nonprofit organizations that are operated, managed and governed by the community with the help of CSOs, local government and committees within the community is essential to have a well-functioning micro grid in Nepal. The Dhapsung Solar Mini-grid community-owned energy projects is one such example, which was installed by Grid Alternatives, operated and managed by the Dhapsung women’s group and supported by DBI.

CASE STUDY I:
DHAPSUNG VILLAGE: RENEWABLE ENERGY SYSTEM MANAGED BY LOCAL WOMEN'S COLLECTIVE

Digo Bikas Institute and its partner have collaborated with the local people of Dhapsung village, Helambu Rural Municipality, Sindhupalchowk district to build a community-owned solar micro-grid. Dhapsung village was devastated by the 2015 earthquake. Since then, community cohesion disintegrated and few communal events, especially those that brought women together, continued.

One of the main challenges for setting-up the community solar micro grid in Dhapsung was the difficult terrain which made access to the village arduous because of the lack of roads reaching the isolated remote village. In addition, villagers of Dhapsung had to carry the power poles, industrial lead acid batteries and solar panels from the nearest road, which is 5 hours from Dhapsung. Key learnings from the Dhapsung community solar micro grid is financial viability of community solar micro grid is possible if there are productive end use customers paying into the micro grid and selling the electricity of solar micro grid to national grid. If the electricity users are only paying for lighting that does not provide enough income to maintain the system over the long-term.

One of the unique features of this community-owned energy project is the setting up of a women's group that is responsible for the management of the community-owned energy project. Nepali society is predominantly patriarchal and most of the leadership and management roles are held by men. Indeed, most of the women of Dhapsung don't have formal education and only few women have attended school to complete their primary education. However, this community-owned solar micro-grid of Dhapsung has broken this dogmatic tradition and put women in the leadership roles. As the women's group is responsible for managing the day-to-day operations of the solar-micro grid, they have the authority to decide suitable tariff rates for the energy that is generated and they decide on effective ways to mobilize the collected tariffs within their community. The women's group consults with the men and other villagers to take inclusive decisions for the proper management of the community-owned energy systems, which has ensured local trust in the project and allowed for increased financial stability. The creation of the women's group to manage the community solar-micro grid has allowed the women of Dhapsung to assemble together for a purpose and to improve their meaningful participation in the decision making process. This has drastically improved the unity amongst the women in the community. Moreover, it has provided them the platform to showcase their leadership abilities.

The Dhapsung solar micro grid energy project can be a pioneer model of a decentralized community-owned energy project that can be replicated across Nepal. It further gives hope to challenge patriarchal practices and the energy systems driven by profit motives. The hope is that we can develop more people-led democratic energy systems to build a more equitable and just future.

The model implemented by the Dhapsung village can and must be replicated in different communities, whereby local people are engaged from the outset of the project design stage. This would, however, require the political will and support from local government both technically and financially. If it were to be implemented on a wider-scale, such interventions should have strong end-use components where small and medium-sized businesses and enterprises should be started that would require and rely on electricity from these solar micro-grid. The Government of Nepal along with donors and supporters would also have a role -- to support the identification of off-grid locations where solar microgrids can be developed, supporting community ownership and full engagement, including opening the space for the critical initial decision of identifying a suitable installation company to come from the communities themselves.

CASE STUDY II:
PROSUMER MODEL: AN EMERGING OPTION TO ADDRESS THE ENERGY CRISIS

The role of households and communities in the energy sector have solely been that of a consumer for a long time. Recently this is changing because individuals at the household and community level have started to produce and self-consume energy from solar micro-grids. The prosumer model is a practice whereby consumers can generate the energy from home-based systems and sell it to the national grid. As per Nepal Electricity Authority's "Photovoltaic Solar Technology Directive 2074", there is a net metering provision. This provision allows a solar system that has a minimum of 500W generation capacity to sell the electricity generated to the national grid. This has been a small positive step to facilitate the prosumer model. The NEA has signed a power purchase agreement to connect solar energy from households to the national grid. However, there have been few successful instances of the prosumer model to date, and thus its effectiveness is yet to be seen or evaluated. Nevertheless, the prospect of introducing the prosumer model option in decentralized renewable energy has many advantages because the model identifies consumers simultaneously as producers and can provide autonomy with regards to people-centered energy generation and consumption.

The prosumer model when adopted in a community setting can help to foster community-based renewable energy initiatives i.e. as a collective prosumer. When individuals become energy prosumers, only the individual might benefit from the access to the renewable energy technology. However, when community owned renewable energy initiatives adopt the collective prosumer model the benefit of the surplus energy generated from the initiative can be reaped by all members of the community ensuring social inclusion. Currently, there is lack of clarity in policy whether it encourages individuals only or also communities to adopt a prosumer model where they can produce energy together at small-scale. However, the prospect of consumers generating the energy and connecting to the national grid could help address the current energy crisis. To date, there is insufficient encouragement from the government at the policy level and a lack of technical support exists for the interested groups who want to connect energy generated from their homes and communities to the national grid. For now, the concept of the prosumer model is new in Nepal, and needs detailed discussion as well as support for its operationalization before it can be effectively implemented.

WAY FORWARD TO EXPAND COMMUNITY-OWNED DEMOCRATIC ENERGY:

Community ownership of renewable energy means that the energy-related assets, such as energy generation systems, energy storage systems, energy efficiency systems etc. can be collectively owned and managed by their users (IRENA, 2020) The energy systems are under the control and ownership of people, not corporations or solar companies. This allows consumers to also become producers, activists acting as social entrepreneurs and citizens that sometimes take on tasks that traditionally have been associated with civil servants (Avelino et al., 2014). To promote community-owned democratic energy systems, the following first steps would need to be taken.

First, the exploitation of various sources of renewable energy (e.g.micro-hydro, mini-solar grids, small-scale biogas) must be prioritized under the control and ownership of communities. The geography of Nepal means such renewable energy sources are readily available in most parts of the country and they must be developed at a small-scales right away to put an end to the energy shortages in areas where energy is not accessible with the current infrastructure. Apart from this, the community must be given ownership and control so that the energy generation, operation and management process becomes decentralized. This would help ensure the energy generation process is locally appropriate, and fostering equity while distributing energy generated from such projects to communities. The involvement of community members also ensures that the energy projects will be sustainable in the long run as the energy generated from such projects will directly impact the livelihoods of the community members as it is tied to their access for reliable and affordable energy.

Second, donor agencies must involve all relevant community stakeholders from the beginning in relation to the planning, financing, operation, maintenance and longevity of renewable energy projects. Moreover, control over the operation of the energy projects must be in the hands of the communities so ensure sustainability while developing a sense of ownership amongst the community members after the donor financing has come to an end.

Third, the sole ownership of renewable energy projects by the private sector must be discouraged. The private sector can be stakeholders in developing renewable energy projects as they bring knowledge and technical expertise for the renewable energy technology development. However, their ownership of the renewable energy process in Nepal has exclusively focused on profit generation currently. The private sector should have a collaborative role in the development of renewable energy projects under the collective ownership of the community with the primary goal of providing affordable and reliable energy and making energy accessible to all.

Lastly, community-owned renewable energy systems must be designed from initiation to take up the collective prosumer model to ensure long-term sustainability of projects that provide reliable energy supplies to the community members. In the present context of Nepal, the community-scaled prosumer model is in its infancy stage, and donor agencies still lack plans to convert community-owned micro grids into the prosumer model from planning to execution phase. However, going forward, the prosumer model can allow community-owned energy systems to be connected to the national or local grid so that the extra energy generated from the system in down-time can be sold. This would allow them to generate revenue from the energy system. The revenue generated could then aid in ensuring the maintenance and sustained operations of the community-owned energy systems.

RECOMMENDATIONS TO BOOST AND OPERATE SUSTAINED COMMUNITY-OWNED PROJECTS:

1. Currently, the national level renewable energy policies such as the National Renewable Energy Framework (NREF) lack participation of all relevant stakeholders especially the involvement of local CSOs and community stakeholders. Their active participation in all of the stages of policy making for renewable energy must be ensured to promote sufficient ownership of community in renewable energy projects.
2. Strict guidelines for donor agencies working in the renewable energy sector should be established to involve all community stakeholders from the very initial designing process of the renewable energy projects and throughout the project cycle, including matters concerning financing, maintenance, operation, longevity and sustainability of any financed energy projects.
3. Policy, legal and institutional frameworks such as renewable energy agencies at all three tiers of government (i.e. local, provincial and federal level) should be established with the sole purpose of increasing the number of community-owned energy projects at the national and sub-national level.
4. Existing obstacles in the form of unfavorable government policies, lack of public resources for community-owned energy initiatives, and limited access to locally appropriate and climate-resilient renewable technologies must be addressed.
5. Government programs with appropriate allocation of funds to technically capacitate local authorities and community members on energy system operation and maintenance, standardization of legal documents, finance management tools and administrative services should be provided (so that community members no longer end up relying on private consultants to manage their community-owned energy systems)
6. Supporting policy amendments should be introduced to ensure fair prices are paid for community-owned renewable energy sold to the national or local grid.
7. Formation of a dedicated team within the government structure to support and enhance community-owned energy projects with the goal of lobbying for community-owned energy systems and removing regulatory barriers that stop its development.
8. The current system of subsidy delivery models for renewable energy projects has grouped community-owned energy projects with co-operative and public private partnership (PPP) energy projects. A new category for subsidy delivery options should be introduced for community-owned energy systems, as they fundamentally differ from co-operative and PPP models of energy systems.
9. Provision of financial incentives such as tax credits and grant funding for community-owned energy systems should be provided by the government, especially for the purposes of battery replacement.
10. Administrative and bureaucratic processes should be simplified and their transparency should be improved, in order to allow for effective engagement of community-owned and operated renewable energy projects.

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